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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations
December - 2025

CS3002-1 - COMPUTER GRAPHICS AND VIUSALIZATION
(For CSE)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. In raster scan displays, a raster is:
A) A mathematical equation
B) A set of scan lines forming a rectangular array of pixels
C) A group of vectors
D) An image filter
2. Which type of display produces smooth lines and is used for line drawing applications?
A) Raster scan display
B) Random scan display
C) LED display
D) LCD display
3. In a true color system, the frame buffer uses:
A) 8 bits per pixel
B) 16 bits per pixel
C) 24 bits per pixel
D) 32 bits per pixel
4. The refresh rate of raster scan displays is typically:
A) 10–20 frames/sec
B) 30–60 frames/sec
C) 100–120 frames/sec
D) 60–80 frames/sec
5. The midpoint circle algorithm is based on the property of:
A) Circle symmetry
B) Polygon convexity
C) Vector interpolation
D) Color interpolation
6. A square of side 2 units has one vertex at (0,0). If scaled by factor 2 in both directions, what will be the side length of the new square?
A) 2
B) 4
C) 6
D) 8
7. In 3D graphics, translation is represented using which type of matrix?
A) 2×2 matrix
B) 3×3 matrix
C) 4×4 matrix
D) 5×5 matrix
8. The composition of transformations means:
A) Applying one transformation at a time only
B) Applying multiple transformations in sequence
C) Undoing a transformation
D) Applying transformation without matrix
9. In perspective projection, the lines of projection:
A) Are parallel
B) Converge at a single point (center of projection)
C) Are perpendicular to the viewing plane
D) Are infinite in number
10. The CMY color model is mainly used in:
A) Monitors
B) Scanners only
C) Cameras
D) Printers



11. The YIQ color model is used in:
 A) Television broadcasting
 B) Printers
 C) Medical imaging
 D) Animation
12. A B-spline curve offers:
 A) Local control of curve shape
 B) No control over shape
 C) Only global control
 D) Random curvature
13. Fractal geometry is mainly used to model:
 A) Perfectly smooth surfaces
 B) Natural objects like mountains or clouds
 C) Mechanical parts
 D) Regular polygons
- In computer graphics, shading models are used to:
 14. A) Represent color models
 B) Reduce rendering time only
 C) Increase object resolution
 D) Determine how light interacts with surfaces
- The Phong shading model improves visual realism by:
 15. A) Ignoring light reflection
 B) Using flat colors only
 C) Calculating lighting per pixel
 D) Displaying wireframe
- Recursive algorithms are often used to generate:
 16. A) Fractal geometry
 B) Bezier surfaces
 C) B-Spline curves
 D) Polygon meshes
- Z-buffer stores which value for each pixel?
 17. A) Color
 B) Depth
 C) Index
 D) Vertex
- Which OpenGL function sets color?
 18. A) glColor()
 B) glVertex()
 C) glMatrix()
 D) glSetColor()
- glBegin() specifies:
 19. A) Transformation
 B) Color
 C) End of drawing
 D) The start of primitive drawing
- A Gasket program demonstrates:
 20. A) Recursive fractals
 B) Vertex arrays
 C) Color attributes
 D) Texture mapping

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain different 2D transformations with suitable diagrams. Discuss how two successive rotations are additive.	8	L*2	2	1
	b) Explain vector display and raster display architecture with suitable diagrams.	8	L2	1	1
2.	a) Explain the data structures used in scan line polygon filling algorithm. Identify the contents of Active Edge Table to fill the polygon with the vertices A (2, 4), B (4, 6) and C (4, 1) for y=1 to y=6.	8	L2	2	1
	b) Write the midpoint circle drawing algorithm and explain 8 way symmetry.	8	L2	2	1
3.	a) Explain the different case involved in Sutherland-Hodgeman clipping algorithm.	8	L2	2	1
	b) Perform a counterclockwise 45 degree rotation of a triangle A(2,3), B(5,5), C(4,3) about point (1,1), Find the resultant matrix and new coordinates.	8	L2	1	1
Unit – II					
4.	a) Discuss Bezier curves and surfaces? Derive the equation for Cubic Bezier curves, where 4 control points are given and degree 3.	8	L2	3	1
	b) Describe Diffuse reflection illumination models.	8	L2	4	1

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| 5. | a) | Define projection? Explain different types of parallel and perspective projections. | 8 | L2 | 3 | 1 |
| | b) | Explain the HSV and YIQ color model with neat diagram. | 4 | L2 | 3 | 1 |
| | c) | Discuss any four B-Spline curves properties. | 4 | L2 | 4 | 1 |
| 6. | a) | Define fractal? Describe its classifications & generation. | 8 | L2 | 3 | 1 |
| | b) | Explain translation and scaling of 3D objects. | 4 | L2 | 2 | 1 |
| | c) | Explain the following Phong shading. | 4 | L2 | 4 | 1 |

Unit – III

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|----|-------|---|---|----|---|---|
| 7. | a) | Explain the following | | | | |
| | (i) | Z- Buffer algorithm | | | | |
| | (ii) | Area subdivision algorithm | 8 | L2 | 4 | 1 |
| | b) | Build a program to recursively subdivide a triangle to form 2D Sierpinski gasket. | 8 | L2 | 5 | 1 |
| 8. | a) | Explain any one list priority algorithm with a suitable diagram. | 8 | L2 | 4 | 1 |
| | b) | Explain the following, with its syntax and an example | | | | |
| | (i) | glBegin() | | | | |
| | (ii) | glutInitDisplayMode() | | | | |
| | (iii) | glutCreateWindow() | | | | |
| | (iv) | glClearColor() | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
